

WANG YAO

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EDUCATION

- 09/2023-10/2024 Faculty of Construction and Environment, Hong Kong Polytechnic University
- Master of Science in Sustainable Urban Development
 - Final GPA 3.51/4.3
- 09/2017-06/2022 College of Civil Engineering and Architecture, Hebei University (HBU)
- Bachelor of Engineering in Architecture
 - A 5-year undergraduate programme with a final GPA of 3.88/5

RESEARCH INTERESTS

Extended Reality, Culture Heritage, Virtual Environmental, Environmental Psychology.

PUBLICATIONS

Jiashuo Cao*; **Yao Wang***; Mark Billingham; Chen Li

Evaluate Impact of Visual Guidance and Biofeedback in VR-supported Breath Training: A Pilot Study; **Siggraph Asia 2025 Technical Communication (Under Review)**

Yufei Lu; Ye Jia; Guang Chen; **Yao Wang**; Peter H. F. Ng; Laura Zhou; Qing Li; Chen Li

Towards Effective Collaborative Learning in Edu-Metaverse: A Study on Learners' Anxiety, Perception, and Behaviour. **Learning Technologies and Systems (ICWL 2024)**

RESEARCH PROJECTS

- 06/2025--present **Agentic Audio Moderator vs Human Moderator in Usability Testing**
- This study investigates the effectiveness of AI agent moderators versus human moderators in facilitating think-aloud usability testing protocols.
 - Literature review, Experimental design and conduction
- 10/2024--present **Evaluate Impact of Visual Guidance and Biofeedback in VR-supported Breath Training**
- The study contributes a novel framework integrating implicit/explicit cues and bio-responsive interaction in VR to significantly enhance box-breathing training efficacy.
 - Unity development, Environment design, Experimental design and conduction, Data Processing
- 12/2024-present **Intercultural Communication Platform in Virtual Platform**
- Foster communication among individuals from diverse cultural backgrounds through collaborative 3D painting tasks in VR.
 - Idea generation, Experimental design and conduction, Data Processing
- 10/2023-07/2024 **PolyU Edu-Metaverse Platform**
- This project develops an Edu-Metaverse platform to deliver interdisciplinary content and explore its potential for active, effective learning.
 - 3D Modelling, Environment planning, Experimental conduction

Other Project

- 11/2024-03/2025 **UI Design for Sensor-enabled Urban Green Care Farm**
- Therapy user interfaces for farming therapy applications, focusing on accessible design principles.
 - UI Design
- 06/2024-08/2024 **The RedMeta Chamber on Spatial.io**
- A Dream of the Red Chamber VR scene offering an immersive experience of Xiaoxiang Guan, with interactive character dialogues and music.
 - Idea generation, Environment design
- 09/2020-06/2021 **Immersive Participation in Ancient Villages' Development in Context of Rural Revitalization: exemplified by Yuanqu Village of Handan**
- This project aims to revitalize Yuanqu Village in Handan through a design-for-equity model.
 - Idea generation, progress management and report drafting
- 09/2019-06/2020 **An App Concept Called "Yigeng Murals": the digital regeneration and activation plan of Taihang Murals**
- Endeavored to help in spreading local culture and protecting Taihang Murals by achieving artistic activation of those murals with adoption of digital technologies.
 - Idea generation and report drafting
- 09/2018-08/2019 **Influence and Enlightenment of How Urban-Rural Economic and Cultural Differences Affect Formation of Middle School Students' Consumption View: A Case Study in Baoding and Cangzhou in Hebei**
- Explored whether and how values and consumption views of middle school students are affected by the urban-rural economic gap.
 - Idea generation and report drafting

WORK & INTERNSHIP

- 07/2024-present **Research Assistant, Department of Computing, PolyU**
- Responsible for modeling and scene construction of virtual environments. Literature collection of human-computer interaction, as well as the development of research ideas.
- 10/2024-03/2025 **Virtual Intern, Australian Research Centre for Interactive and Virtual Environments (IVE)**
- Collaborate on the development of the virtual environment for VR breathing training projects.
 - Proposal drafting, conducting experiments, data collecting and analyzing
- 05/2023-08/2023 **Intern, Shenzhen Urban Planning & Land Resource Research Center**
- Participated in several important planning or policies for Shenzhen, working as an assistant Urban and Rural Planner.
 - Report drafting, team cooperation
- 06/2022-08/2022 **Game Art Designer, Beijing Bilibili Yuanli Xingju Network Tech Co., Ltd**
- Responsible for game module integration, map building and iteration, and model data processing
 - Stylized scene construction, data analytics and overall planning, and spatial processing and 3D modelling

ACTIVITIES

09/2017-06/2020	President, Taihang Local Culture Research Society ➤ Led peers to carry out various activities, e.g. field trips and photo exhibitions, to investigate or promote local cultures.
09/2019	Surveyor, Ancient Buildings Mapping & Protection in Yuanqu Village, Handan ➤ Surveyed and mapped the villages dating back to the Ming and Qing Dynasties ➤ Sorted out and summarized the data collected
10/2019	Surveyor, Mapping and Research Project related to Baoding Industrial Heritage ➤ Involved in mapping and data collection of the old plant site

AWARDS

Commendation Award (2024)	<i>PolyU Metaverse Contest 2023/24</i>
Provincial-Level Grant (2021)	<i>National College Student Innovation & Entrepreneurship Training Program</i>
Provincial Third Prize (2020)	<i>"Chuang Qingchun" National College Student Innovation & Entrepreneurship Competition</i>
University Scholarships (2019,2020,2021)	<i>Second Prize & Third Prize (twice)</i>

PROFICIENCY & SKILLS

Language	➤ IELTS overall 7.0
Software	➤ Unity, Blender, Figma, Photoshop, Nvivo, SPSS, Stata, AutoCAD, Arcgis
Photography	➤ A signed contributor to tuchong.com, an online community of photographers
Writing Expression	➤ Posted poems of over 10,000 words on online platform for literature creators
Fine Art	➤ Aquarelle, Mark Painting, sketch